CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 4: Transaction Concurrency Control Protocols

Course: Advanced Database Engineering [23CS3503]
Category: Assignments | Assignment #4 | Academic Year: 2026-27

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1. Overview & Objective:

ACID guarantees, conflict serializability, locking protocols, and deadlock resolution.

2. Assignment Problems & Instructions:

1. Determine if a given transaction schedule is conflict serializable using precedence graphs.
2. Prove that Strict Two-Phase Locking (2PL) guarantees conflict serializability without cascading aborts.
3. Explain Wait-Die and Wound-Wait deadlock prevention schemes based on timestamps.
4. Compare Multiversion Concurrency Control (MVCC) with lock-based concurrency control.

Submission Guidelines & Verification:

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